INTRODUCTION

Hamilton Marketing Services is a small marketing consulting firm that provides a wide range of marketing and advertising services to small to medium-sized retail firms. Recently, a pet-grooming company has sought expert input from Alex, the founder of Hamilton Marketing Services. This dog-grooming company is considering changing the pricing structure for its full-service dog-grooming service. The company is considering two pricing options. The first pricing option is a flat rate of $40.00 per visit, while the second is $30.00 per visit if the dog owner signs up for four dog groomings. To identify the most effective pricing strategy, Alex decided to send out fliers to two random samples of 80 customers. Each flier included an attached coupon. One sample of customers would receive a coupon for the $40 price, while the other sample would receive a coupon for the $30 four-visit package. Alex has tasked our team with constructing a 95% confidence interval for the difference in response proportions and determining which pricing strategy is preferred.

1. PROBLEM STATEMENT AND ASSUMPTIONS

The purpose of this report is to evaluate the difference in response proportions for a pet-grooming company seeking services from Hamilton Marketing Services. This analysis is based on data collected from two random samples of 80 customers who received promotional fliers offering two different pricing strategies. Our assumptions include: each sample of 80 customers is random and independent, survey response matches purchase behavior, revenue calculations assume each visit is paid in full, and pricing changes remain flat. With this, our team will construct a confidence interval for the difference in response proportions to help determine if the difference in response rates is statistically significant. Additionally, we will compare the total revenue generated under each pricing strategy to determine the most lucrative option for the pet-grooming company. This report will calculate a 95% confidence interval for the difference in response proportions, determine the sample proportions for each coupon option, compare the revenue generated by the two options, and evaluate additional factors or issues to consider when deciding which coupon options to use. As a result of this report, Alex will be able to make a final pricing recommendation supported by data-driven guidance.

2. SUMMARY OF RESULTS

The $40 coupon yielded a 20% response rate, with 16 of 80 individuals indicating they would be interested in the $40 per visit grooming service. The $30 four-visit coupon produced a 17.5%

response rate, with 14 out of 80 individuals indicating they would be interested in the $ 30 per visit, four-visit grooming service. The 95% confidence interval for the difference is [-0.0959, 0.1459], indicating that the true difference in response rates between the two pricing options could range from –9.59 to +14.59 percentage points. Since the confidence interval crosses zero, there is no statistically significant difference in the response rates between the two pricing options. The $30 for four visits pricing option may generate a higher observed revenue than the $40 per visit option ($1,680 vs. $640). The qualitative analysis identified several risks and challenges associated with both the quantitative assessment and the pricing options. Key risks identified in this analysis include the risk that customers do not complete all four visits, the potential hesitation on the part of both the groomer and the customer to enter into the four-visit agreement up front, and the administrative burden that may accompany the second pricing option. Limitations of the study include the short one-month response window, the absence of competitive benchmarking, and the lack of insight into why customers responded as they did. The identified qualitative factors do not disqualify either option but do inform implementation.

3. ANALYSIS

Quantitative

Pricing Test Overview

The $40 coupon produced a response rate of 20.0% (16 of 80). The $30 four-visit coupon produced a response rate of 17.5% (14 of 80) (Table 1).

Coupon Response Comparison			
Offer	**Responses (Yes)**	**Mailed**	**Sample Proportion ($\hat{p}$)**
$40	16	80	0.2000 (20.00%)
$30 × 4	14	80	0.1750 (17.50%)

Table1: Coupon responses comparison for $40 and $30x4 options.

Response Rate Analysis

A 95% confidence interval was calculated to estimate the difference in response rates between the $40 coupon ($p_1$) and the $30 per visit with four-visit commitment coupon (p_2). The observed difference in sample proportions was 0.025 (Table 2). Using the normal approximation method, the 95% confidence interval for the difference ($p_1 - p_2$) is (−0.0959, 0.1459) (Figure 1).

This means the true difference in response rates could reasonably fall between −9.59% points and 14.59% points (Table 2).

Difference in Response Rates	
Measure	**Value**
Difference ($\hat{p}_1 - \hat{p}_2$)	0.025
Standard Error	0.06168
95% Confidence Interval Lower Bound	−0.0959
95% Confidence Interval Upper Bound	0.1459

Table 2: Difference in response rates data

The 95% confidence interval for the difference in response rates ($p_1 - p_2$) is (−0.0959, 0.1459). Since this interval includes 0, there is no statistical evidence at the 5% level to conclude that the response rates differ between the two coupon options. The observed difference in the sample appears to be due to random variation rather than a meaningful difference in customer response. There is a meaningful difference in revenue between the two coupon options.

Revenue Analysis

The $40 offer generated 16 responses, resulting in total revenue of $640 (16 × $40). This equals $8.00 in revenue per mailed coupon (Table 3).

The $30 per visit option requiring a four-visit commitment generated 14 responses. At full completion of the four visits, each responding customer produces $120 in revenue, resulting in total revenue of $1,680 (14 × $120). This equals $21.00 in revenue per mailed coupon (Table 3).

The sensitivity table shows how revenue changes as the completion rate for the four-visit package varies (Table 4). If fewer than 38.1% of the package value is realized, the $40 option produces more revenue. At 38.1%, both options generate the same revenue (Figure 1). Once completion rises above that level, the $30 × 4 option becomes the stronger financial choice, and the revenue advantage increases as completion improves.

This confirms that the four-visit package does not need full completion to outperform the single visit price. Based on the data, the $30 × 4 pricing strategy generates more revenue than the $40 single visit option.

Offer	Yes Answers	Mailed	Price	Series Value	Completion Rate	ERPM	Observed Revenue	Break Even Completion Rate	Current winner
$40	16	80	$ 40.00	0	100%	$ 8.00	$ 640.00	38.10%	$30x4
$30×4	14	80	$ 30.00	$ 120.00	100%	$ 21.00	$ 1,680.00		

Table 3: Revenue and Break-Even Analysis by Pricing Option

$$Break\ even\ completion\ rate = \frac{\$1680}{\$640} = 38.1\%$$

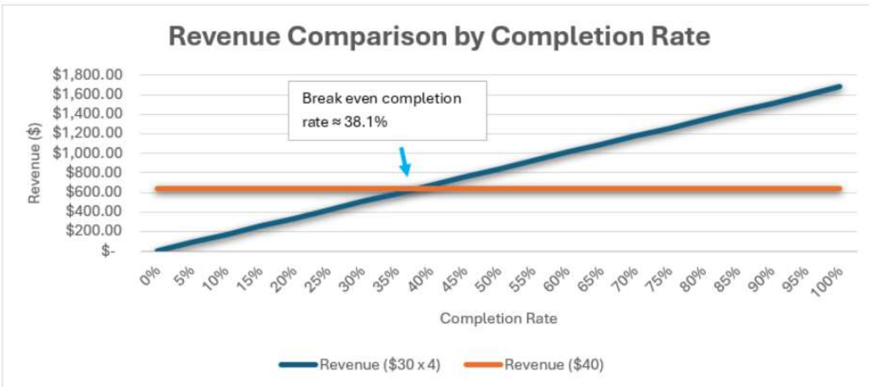

Figure 1: Revenue comparison by completion rate graph

Completion Rate	Revenue ($30×4)	Winner
25%	$420	$40
38.10%	$640	Equal
50%	$840	$30×4
75%	$1,260	$30×4
100%	$1,680	$30×4

Table 4: Sensitivity analysis for break-even points

Qualitative

Several qualitative factors should be considered when evaluating the two pricing options. First, the survey collected data from potential customers, not existing customers. Existing customers may react differently, particularly long-standing customers who expect to pay $40 per grooming session. Adverse reactions reported by existing customers are not included in this analysis. Additionally, customers may often want a trial run to see if they like the groomer before committing to a series of groomings. The groomer may also want a trial run to see if they can work with the pet, as some are aggressive or heavily matted, which increases difficulty for the same rate. Locking into a series of grooming sessions up front may give both parties reason to hesitate.

The second option, four groomings at $30, requires either that the customer returns three times after the initial visit without going to another groomer, or that the customer pays up front for all four sessions. In the first scenario, there is nothing to guarantee the customer will return, which poses a risk of missed revenue. In the second scenario, the customer may be hesitant to pay a large amount up front, especially if they visit the groomer only 2 to 4 times per year, meaning they could be paying 1 to 2 years in advance. From the groomer's perspective, operating costs can significantly fluctuate over that timeframe, and if the price is locked in, they may be providing grooming services at a loss if costs increase.

Option two also creates additional administrative burden. Tracking the amount of grooming's per customer and explaining the pricing structure to new customers both take time away from grooming, which is detrimental to a business whose revenue is directly connected to the number of grooming sessions completed. More time spent on administrative tasks is less time spent grooming.

Several limitations of the analysis itself are also worth noting. The one-month period to collect responses may have been too short, as many dogs will not immediately need grooming within that window, which could have suppressed responses to both options, particularly option 2. The flat rate structure is also not sensitive to variability in dog breed and size, as a large dog will take considerably longer and cost more to groom than a small one. Competitive benchmarking was not included in this analysis; it is unknown what competitors charge per grooming, what their pricing strategies are, or what the current level of competition is in the local market. This analysis also does not consider whether one pricing structure is more likely to generate repeat customers, though option two arguably creates a stronger incentive for return visits. Finally, the survey does not capture the context behind participants' responses, which is a missed opportunity to better understand customer thinking.

4. RECOMMENDATION

It is recommended that Alex guide the pet grooming company to choose the second option, with a $30 per visit cost with a four-visit series. Since there is no statistically significant difference in the survey results for the two pricing options, the revenue projection and qualitative analysis are most important in guiding Alex's recommendation to the pet grooming company. The revenue projection suggests that the $30 series of grooming services will have better observed revenue. About 2 in 5 customers would need to complete all four services to exceed the revenue projection for the $40 flat rate. The qualitative assessment identifies risks and limitations associated with the assessment and pricing option two. These aren't necessarily reasons not to pursue option two. Rather, they will help inform how to successfully implement option two. To mitigate the completion risk, the grooming company can ask for payment for all four visits up front ($120). Requiring customers to pay for the series up front eliminates the administrative burden that could be introduced if payments were made per visit. It is recommended to confirm the acceptability of this approach through customer input before implementation. The current survey information does not necessarily inform willingness to pay in advance. The grooming company may consider only offering the $30 four visit series to repeat customers who have completed a full price visit, not first-time customers. This allows the groomers and customers to vet each other first. Customers can decide if they like the groomer, and the groomer can decide if they like the customer. The relationship can then evolve from the initial visit into a long term one. To mitigate the risk of existing customer backlash, the organization can allow existing customers to continue at the $40 rate and allow opt in to the second pricing strategy. To mitigate the risk of increasing operating costs over time (such that $30 fee is no longer sufficient to cover costs and achieve desired profit margin), a time limit can be set on the four-visit series so that it is not indefinite. Effective communication of the second pricing alternative will be essential to achieve successful adoption. The second pricing alternative should be promoted as a prepaid package with a discount: pay now to save $10 per visit.

5. ACTION PLAN

We organized the action plan into three phases: validation, implementation, and monitoring. Laying it out this way makes the steps easier to follow and helps the groomer clearly see how the plan should unfold over time and whether it's working as expected.

Phase 1: Validation

The pet grooming company should move forward with the $30 × 4 pricing option, provided customers pay for the full package upfront and complete all four visits within a year of purchase. To proceed with the implementation of this pricing strategy, the dog grooming company should retain the $40 per visit price for existing and new customers. Prior to implementing the new $30 for four visit program, the company should consider collecting feedback on the willingness to pay up front for the four visits since the current data does not capture the willingness to pay up front. Additionally, they should conduct a competitor analysis to confirm that this proposed pricing structure is acceptable. The cost structure should be evaluated alongside revenue. Labor, supplies, and other variable costs per grooming visit should be reviewed to confirm that the four-visit package improves overall profitability rather than simply increasing volume.

Phase 2: Implementation

They should allow existing customers who have completed at least one $40 visit, the opportunity to opt in to the $30 for four visit program. The company should collect payment for the four $30 visits up front when the customers enroll in the program. To avoid revenue and scheduling uncertainty, the package should require that all visits be used within 12 months of purchase. The company will need to track the one-year expiration date on the four-visit series. Once the willingness to pay up front is confirmed, the grooming company should begin promoting the new pricing program, which requires payment up front for four visits to save $10 per visit. In the future, the grooming company can consider implementing tiered pricing based on the dog's size to better account for size variability, which impacts the length of the visit and the total cost to provide the service.

Phase 3: Monitoring

After launch, completion rates should be tracked. Management should monitor how many customers complete all four visits and how quickly they move through the schedule. Comparing the actual completion rate to the 38.1% break-even threshold will confirm whether the expected revenue advantage is being realized. Operational capacity should also be reviewed as repeat visits increase. Appointment availability, staff workload, and service quality should be monitored to ensure that the higher visit frequency does not create scheduling bottlenecks or reduce the customer experience. Growth in visits is beneficial only if it can be supported operationally. Finally, customer retention beyond the four-visit series should be measured. If customers continue booking appointments after completing the package, the long-term value of the program may exceed the initial revenue comparison. After several months of data collection, performance should be reassessed, and adjustments made as needed.